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# =========================
# 1. Import Libraries
# =========================
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score
from scipy.cluster.hierarchy import dendrogram, linkage

# =========================
# 2. Load Dataset (UCI Wine)
# =========================
wine = load_wine()
X = pd.DataFrame(wine.data, columns=wine.feature_names)
y = wine.target # Keep target for potential future use or reference

# =========================
# 3. Standardization
# =========================
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# =========================
# 4. Find Best K (Elbow + Silhouette)
# =========================
inertia = []
silhouette = []
K_range = range(2, 10)

for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_scaled)

    inertia.append(kmeans.inertia_)
    silhouette.append(silhouette_score(X_scaled, labels))

# Plot Elbow
plt.plot(K_range, inertia, marker='o')
plt.title("Elbow Method")
plt.xlabel("K")
plt.ylabel("Inertia")
plt.show()

# Plot Silhouette
plt.plot(K_range, silhouette, marker='o')
plt.title("Silhouette Score")
plt.xlabel("K")
plt.ylabel("Score")
plt.show()

best_k = K_range[np.argmax(silhouette)]
print("Best K:", best_k)

# =========================
# 5. K-Means WITHOUT Standardization
# =========================
kmeans_raw = KMeans(n_clusters=best_k, random_state=42, n_init=10)
labels_raw = kmeans_raw.fit_predict(X)

score_raw = silhouette_score(X, labels_raw)
print("KMeans (No Scaling) Silhouette:", score_raw)

# =========================
# 6. K-Means WITH Standardization
# =========================
kmeans_scaled = KMeans(n_clusters=best_k, random_state=42, n_init=10)
labels_scaled = kmeans_scaled.fit_predict(X_scaled)

score_scaled = silhouette_score(X_scaled, labels_scaled)
print("KMeans (Scaled) Silhouette:", score_scaled)
```

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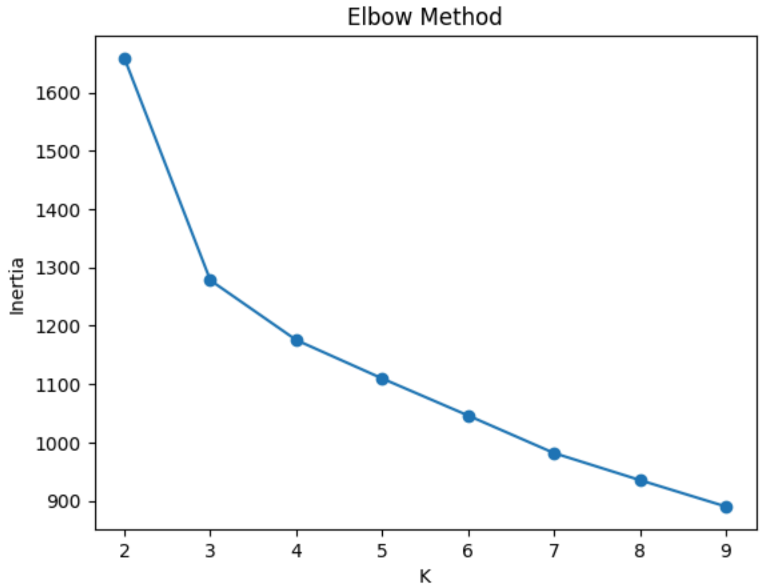
# ========================
# 7. Hierarchical Clustering
# ========================
# Dendrogram
linked = linkage(X_scaled, method='ward')

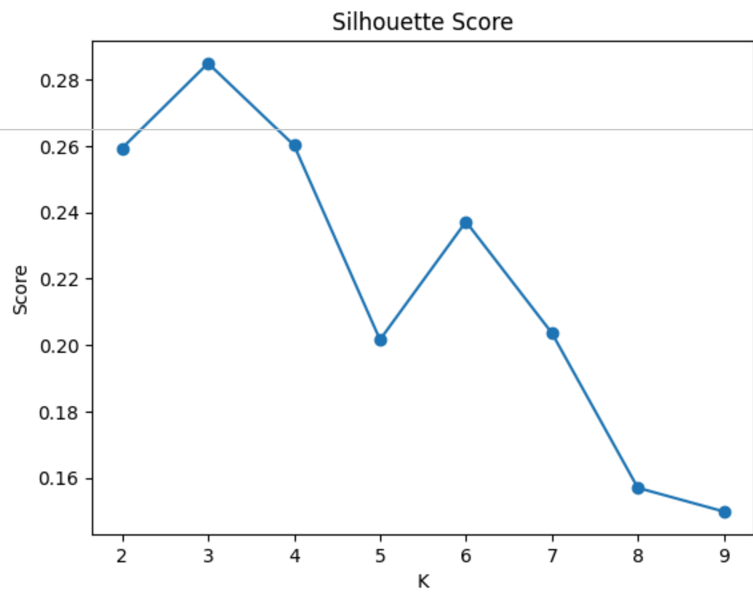
plt.figure(figsize=(10, 5))
dendrogram(linked)
plt.title("Hierarchical Clustering Dendrogram")
plt.show()

# Apply Agglomerative Clustering
hc = AgglomerativeClustering(n_clusters=best_k)
labels_hc = hc.fit_predict(X_scaled)

score_hc = silhouette_score(X_scaled, labels_hc)
print("Hierarchical Clustering Silhouette:", score_hc)

# ========================
# 8. Comparison Summary
# ========================
print("\n--- Comparison ---")
print(f"KMeans (No Scaling): {score_raw:.4f}")
print(f"KMeans (Scaled):     {score_scaled:.4f}")
print(f"Hierarchical:        {score_hc:.4f}")
```



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Best K: 3
KMeans (No Scaling) Silhouette: 0.5711381937868838
KMeans (Scaled) Silhouette: 0.2848589191898987
```